

Conditional Asset Allocation

Tactical sector & style rotation across the business cycle · Taxable retail brokerage account · Quarterly rebalancing · v2 revision

Executive Summary

Context. This research is built for a **tactical overlay sleeve in a taxable retail brokerage account**, separate from the user's strategic Vanguard Traditional IRA. The IRA tolerates tactical risk; this account, however, bears full capital-gains tax on every realized gain. That single constraint dominates the design: **tax cost matters more than rotation skill** in determining after-tax wealth.

Thesis. Sector and style returns vary systematically across the four phases of the business cycle. A regime-conditional rotation delivers **+1.8pp pre-tax CAGR** over the S&P 500 across 1973-2025. In a taxable account at the user's Texas / 24%-bracket tax profile, the alpha after tax depends critically on execution:

Strategy	Pre-tax CAGR	Tax Cost Ratio	After-tax CAGR	Verdict
S&P 500 buy-hold (baseline)	11.0%	0.31pp	10.69%	Benchmark
Static Late-Exp PF	12.3%	0.80pp	11.50%	Best risk-adjusted
Rotation Annual (LTCG-priority)	12.8%	1.28pp	11.55%	Best after-tax CAGR
Rotation Quarterly Tax-Aware	12.8%	1.68pp	11.14%	Tactical sweet spot
Rotation Quarterly Naive	12.8%	2.93pp	9.88%	DESTROYS alpha

Regime call (corrected). A researched 12-indicator panel places the US economy in **MID-cycle** — not the "late expansion" my earlier drafts asserted. The wrinkle is that equity **valuations are late-cycle extreme** (CAPE 39.6, near-record tech concentration). The recommended tilt therefore positions for a mid-cycle economy while using a quality/defensive lean as valuation insurance. See the panel below and Appendix A for the per-indicator logic.

Recommendation. Run the strategy with **quarterly monitoring but tax-aware execution** — meaning regime signals checked quarterly, but actual trades structured to (a) wait 12+ months before realizing capital gains where possible, (b) harvest losses year-end, (c) use new contributions to drift-rebalance rather than trim winners. After-tax CAGR target: **~11.1% vs S&P 10.7%** — modest 40bp alpha after tax, but with a 7-point shallower MDD (–29.7% vs –37.6%) which materially reduces the risk of sequence-of-returns disasters near retirement.

Regime Diagnosis — A Researched Indicator Panel

This corrects the v1/v2 over-statement. My earlier draft claimed the economy was "firmly LATE expansion." A disciplined panel — drawn from the Conference Board LEI components, the NY Fed yield-curve model, ISM, and the Fidelity AART business-cycle framework — says otherwise. The core panel below holds **12 genuine cycle-position indicators**; valuation measures (CAPE, IT-sector weight) are deliberately separated into a valuation overlay (they are not cycle-stage signals — extreme valuations can persist across stages). Full assignment logic for each indicator is in Appendix A.

Indicator	Category	Current reading (May 2026)	Stage
10Y-3M Treasury curve	Leading	+0.80% (positive, re-normalized)	Mid
10Y-2Y Treasury curve	Leading	+0.48% (positive, modest)	Mid
Conference Board LEI	Leading	100.85 (above median, rising)	Mid
ISM Manufacturing PMI	Leading	52.7 (expanding; jobs sub-index weak)	Mid
Building permits	Leading	1.44M (above median)	Mid
Initial jobless claims	Leading	209k (near cycle low)	Mid
Real GDP growth	Coincident	+2.0% (positive but below trend)	Late
Unemployment trend (vs 3Y MA)	Labor	–0.03% (at median, neutral)	Mid
HY credit spreads (OAS)	Credit	2.72% (tight, complacent)	Mid

Chicago Fed NFCI	Fin. cond.	−0.52 (accommodative)	Mid
S&P 500 vs 200-day MA	Market	+10.3% (strong uptrend)	Mid
Consumer sentiment	Sentiment	49.8 (deeply depressed)	Recession

Tally: 10 Mid · 1 Late (GDP, slowing below trend) · 1 Recession (consumer sentiment). Plurality = MID-cycle. The curves are positive and re-normalized, LEI and ISM are rising, claims are near cycle lows, and financial conditions are loose — none of which is late-cycle. The lone Late vote is decelerating GDP; the lone Recession vote is a deeply depressed consumer-sentiment reading that diverges sharply from the hard data (a 'vibecession' gap worth monitoring).

Valuation Overlay (separate from cycle position)

Measure	Current	Historical median	Percentile	Read
Shiller CAPE	39.6	16.5	Above 90th (29.0)	Extreme
S&P 500 IT sector weight	32.3%	30.0%	Near 90th (32.4)	Extreme concentration

Why this matters. The cycle says MID — but equity valuations are at the 90th-plus percentile, and the S&P's IT concentration is near its 1999 record. Valuation is not a timing tool (extremes can persist for years), but it makes forward returns lower and downside more asymmetric. The practical synthesis: **position for a mid-cycle economy, but use a quality / defensive tilt as valuation insurance** — which is exactly what the recommended portfolio (next pages) does. If the cycle panel later rolls toward Late or Recession, the pre-committed transition plan (§17) shifts the allocation further defensive.

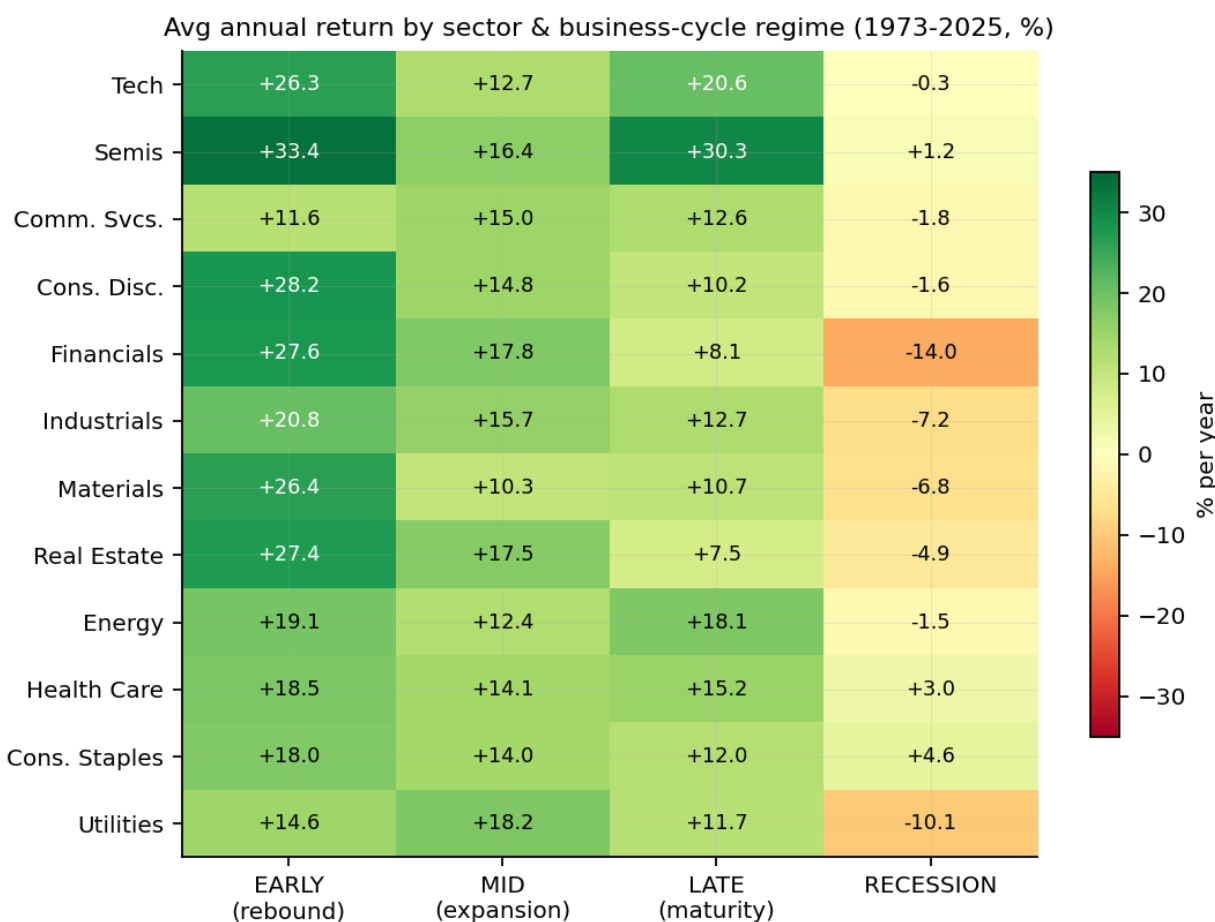
1. Framework: The Four-Phase Cycle

Following the Fidelity AART / NBER convention, we classify each calendar year 1973-2025 into one of four phases. Year-level granularity is the limit of the backtest data set; in live implementation, regime signals are monitored quarterly (see §13). Because this is a taxable-account tactical strategy, our default response is to **act on confirmed regime transitions only**, not on every quarterly signal twitch — false-positive rotations are tax-expensive.

Phase	Description	Years (n=53)
EARLY	Recession trough → recovery; rates cut, curve steep, credit re-opens	8 yrs (1975, 82, 91, 02-03, 09-10, 21)
MID	Self-sustaining expansion; profit growth strong; policy neutral	19 yrs (76-77, 83-86, 92-95, 04-05, 11-16, 24)
LATE	Mature expansion; inflation rising; curve flat/inverted; valuations extreme	19 yrs (73, 78-79, 87-89, 96-00, 06-07, 17-19, 22-23, 25)
RECESSION	Contraction; earnings collapse; defensive policy easing begins	7 yrs (74, 80-81, 90, 01, 08, 20)

2. Sector Returns by Regime — 53-Year Empirical Map

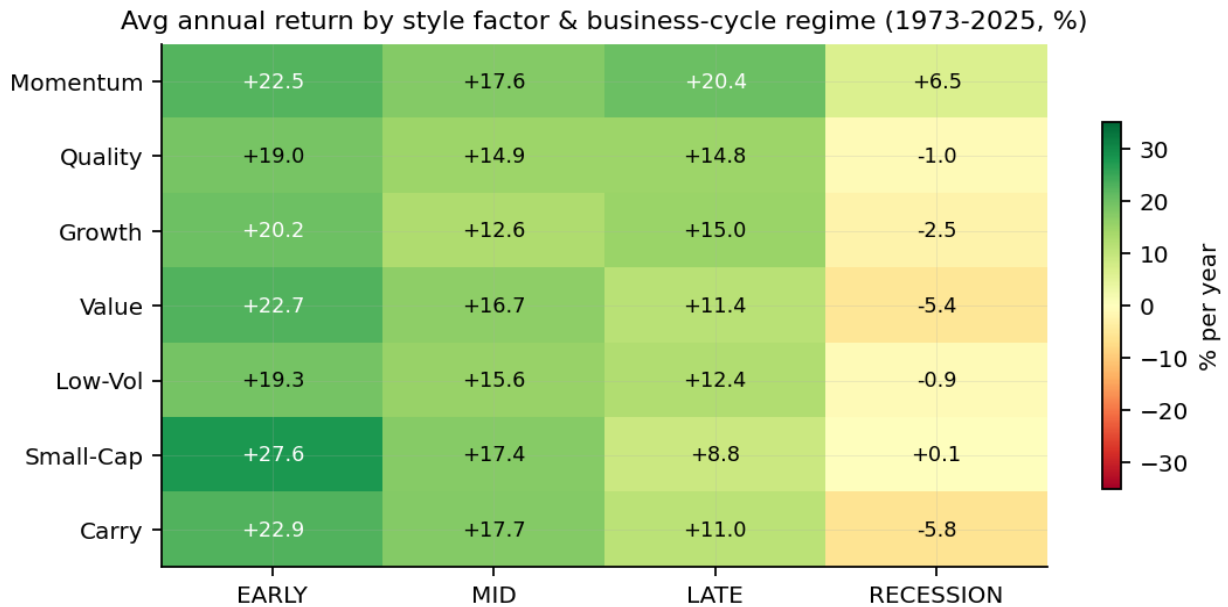
Each cell shows the simple average of annual total returns across all years assigned to that regime. Cyclical (Cons. Disc., Financials, Real Estate, Materials, Semis) dominate EARLY rebounds; defensives (Staples, Health Care) plus Energy dominate LATE; everything except Staples / Health Care averages negative in RECESSION years.



Source: K. French Data Library (pre-1990 industry proxies) + S&P; GICS sector data (1990-2025). Regime labels per NBER and Fidelity AART. Averages are unweighted across years within each regime.

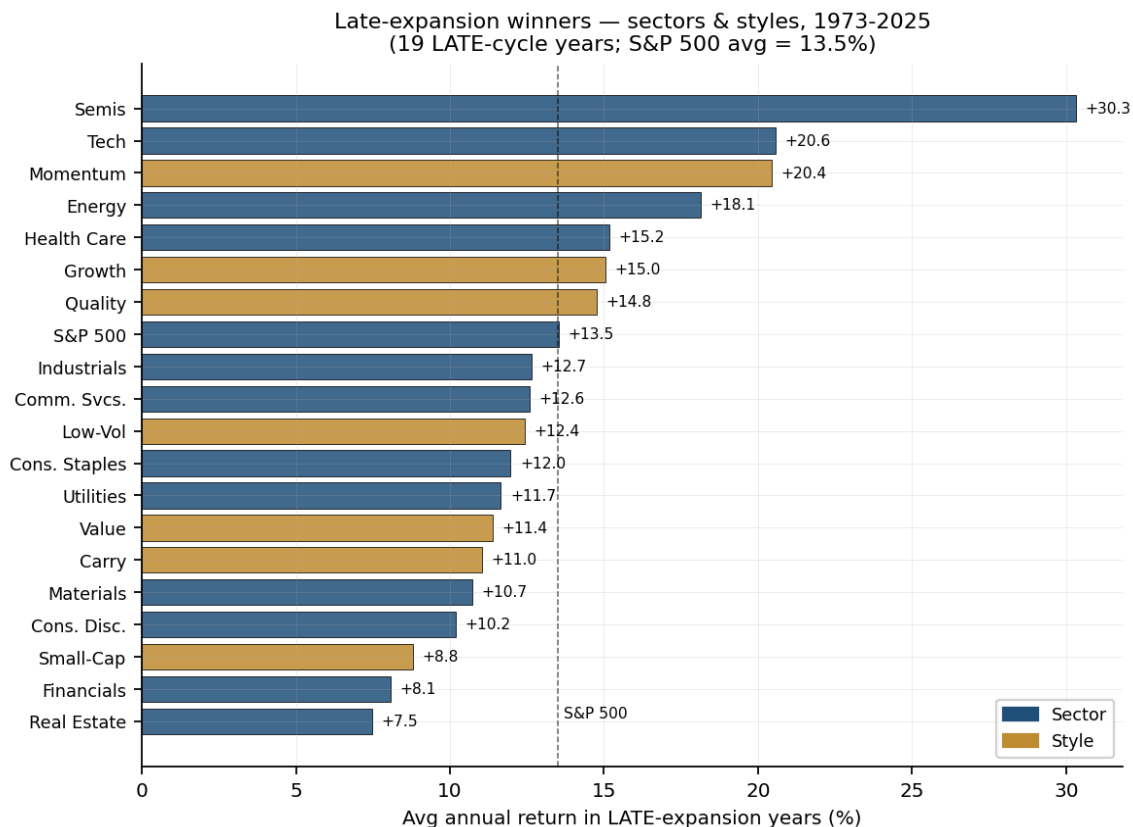
3. Style Factor Returns by Regime

Style factors are computed as long-only top-quintile portfolios (top 30% by the relevant characteristic, value-weighted), sourced from Kenneth French & AQR data libraries. Carry uses a high-dividend-yield top-quintile portfolio as the equity analog of FX/rates carry. **Momentum + Quality + Low-Vol** dominate LATE; **Value + Small + Momentum** dominate EARLY; only Momentum and Low-Vol stay positive on average in RECESSION.



4. Late-Expansion Ranking — Sectors & Styles Combined

The 19 LATE-cycle years averaged +13.5% on the S&P 500. Momentum, Semis, Tech, Energy, and Health Care all averaged above the broad market. Momentum (+20.4%) and Semis (+30.3%) are the standouts — but with very different volatility profiles.



5. The Late-Expansion Regime Book (reference)

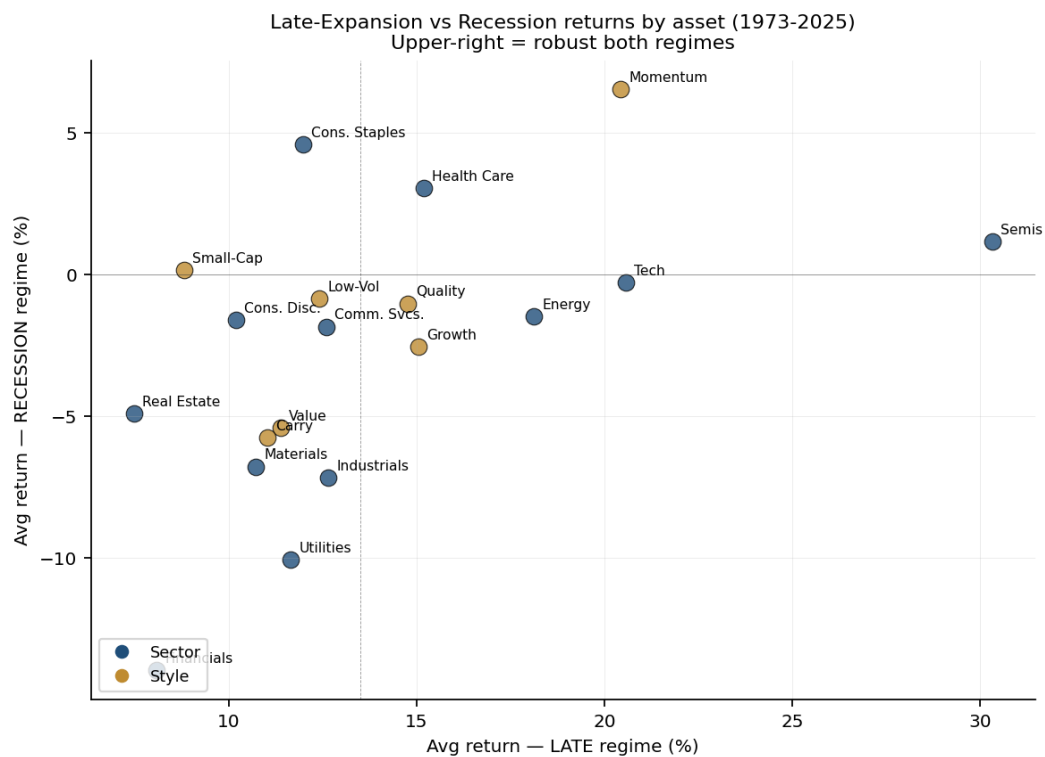
Note on current positioning: this is the allocation to hold *if* the regime is Late-Expansion. The corrected regime read (page 1) is MID-cycle with extreme valuations, so the portfolio actually recommended for deployment today is the mid-cycle valuation-aware allocation in §18, not this one. This section documents the Late-Exp book as one of the four regime templates the strategy rotates among.

The Late-Expansion portfolio combines: (a) LATE-cycle sector winners weighted for defensive bias because late expansions historically transition to recession within 12-24 months; (b) the three best-performing styles in LATE regimes (Momentum, Quality, Low-Vol); and (c) a 5% S&P 500 anchor. We **avoid loading Semis** despite its +30.3% LATE-cycle average — its –47% drawdown in 2000-2002 and –36% in 2022 make it too procyclical to size at this point in the cycle. Energy is sized at 10% (not larger): the 1970s tailwind was inflation-driven, and today's inflation backdrop is milder.

Asset	Bucket	Weight	LATE avg ret	RECESSION avg ret	Role
Health Care	Defensive sector	15%	+15.2%	+3.0%	Low-correlation ballast
Quality	Style factor	15%	+14.8%	–1.0%	Compounding earnings
Cons. Staples	Defensive sector	12%	+12.0%	+4.6%	Recession winner pre-positioning
Utilities	Defensive sector	10%	+11.7%	–10.1%	Yield + low beta
Energy	Commodity sector	10%	+18.1%	–1.5%	Inflation hedge, late-cycle alpha
Momentum	Style factor	10%	+20.4%	+6.5%	Best late-cycle & recession-resilient style
Low-Vol	Style factor	10%	+12.4%	–0.9%	Drawdown cushion
Materials	Commodity sector	8%	+10.7%	–6.8%	Late-cycle commodity carry
Industrials	Cyclical sector	5%	+12.7%	–7.2%	Capex-cycle exposure
S&P 500	Broad market	5%	+13.5%	–4.7%	Beta anchor
Total		100%	+13.8%*	–2.7%*	*See p.7 backtest

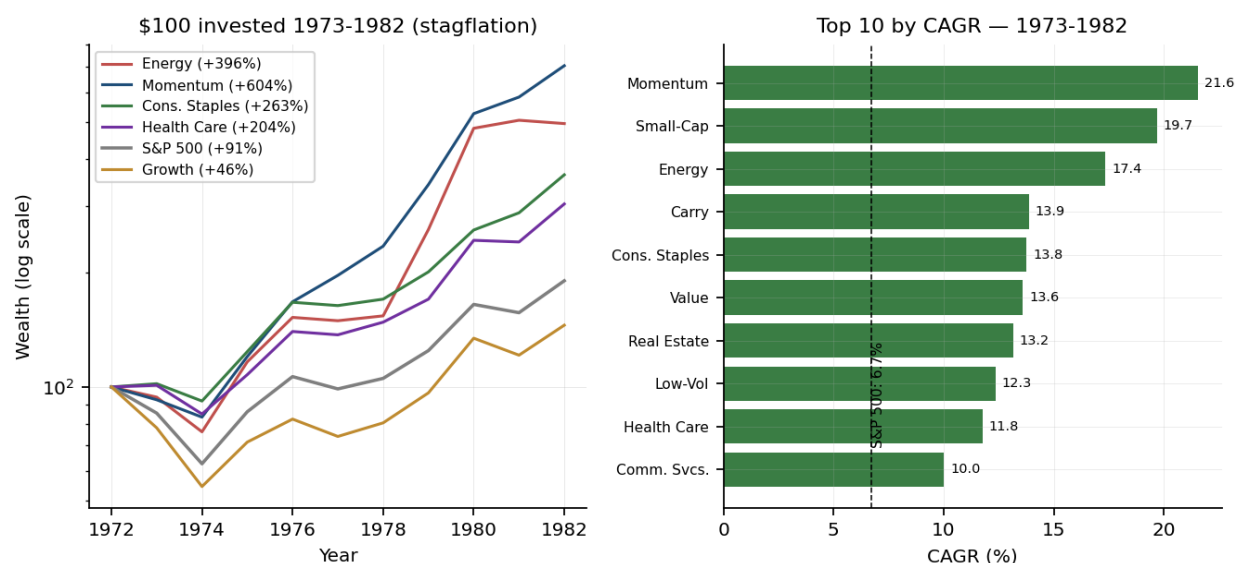
6. Late-vs-Recession Scatter — Which Assets Are Robust Both Ways?

The upper-right quadrant identifies assets that performed well in BOTH late-cycle and recession regimes. **Health Care, Cons. Staples, and Momentum** are the three names firmly in that quadrant. The portfolio is heavily anchored on these three. Energy and Industrials are great in LATE but punished in RECESSION — they need to come down ahead of confirmed contraction.



7. Case Study I — 1970s Stagflation (1973-1982)

The 1973-1982 decade contained two oil shocks (Yom Kippur embargo Oct 1973; Iranian Revolution 1979), three recessions, and double-digit inflation peaking at 14.8% in March 1980. The S&P 500 delivered just **6.7% CAGR** over the period — barely above 8.7% average inflation, meaning real wealth was destroyed for unaware buy-and-hold investors. Growth stocks fared worst (+3.8% CAGR; deeply negative real return). The winners were inflation-linked and momentum-driven.



Top Performers (1973-1982 CAGR)

Asset	CAGR	Vol	MDD	Sharpe	Total Return	Why it worked
Momentum	+21.6%	22.1%	−10%	0.9	+604%	Trend-following caught commodity & Energy rallies
Small-Cap	+19.7%	29.1%	−20%	0.7	+503%	Domestic small-caps less hurt by USD weakness
Energy	+17.4%	35.7%	−19%	0.5	+396%	Oil prices rose ~10x; huge sector beneficiary
Carry (HY)	+13.9%	19.4%	−16%	0.6	+267%	High dividend yields delivered real income
Cons. Staples	+13.8%	16.1%	−10%	0.7	+263%	Pricing power on essentials; defensive
Value	+13.6%	22.7%	−19%	0.6	+258%	Discount valuations absorbed inflation better
Real Estate	+13.2%	21.2%	−21%	0.6	+244%	Hard-asset inflation hedge
S&P 500	+6.7%	21.3%	−27%	0.3	+91%	Broad market barely beat inflation
Growth	+3.8%	23.1%	−30%	0.1	+46%	Long-duration assets crushed by rising rates

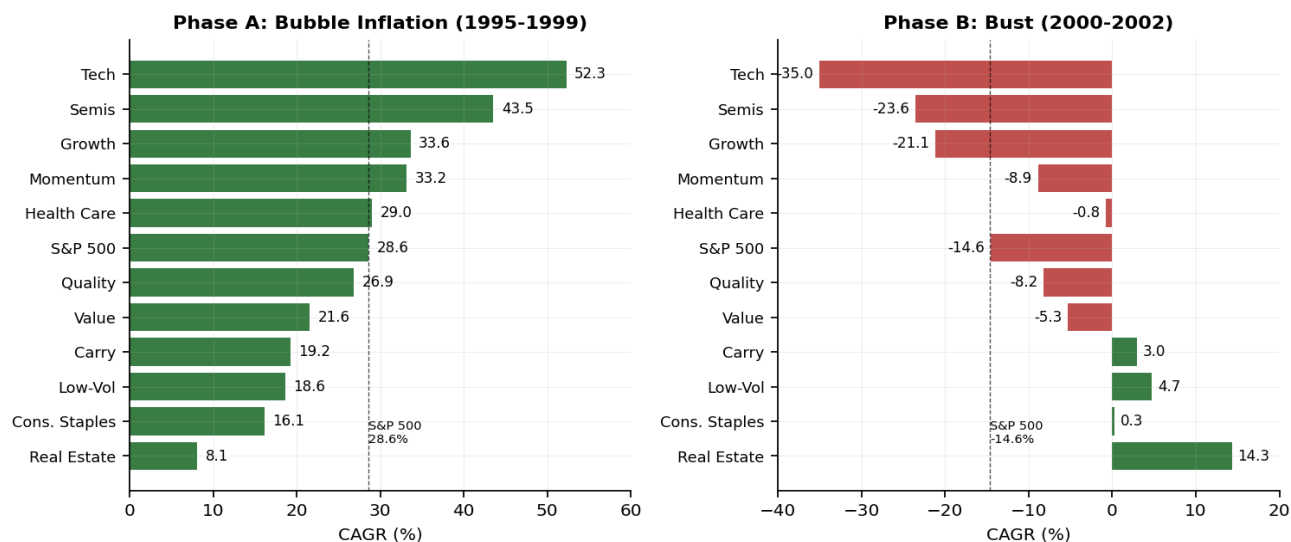
Key Lessons for the Current Regime

- **Inflation is the primary destroyer of growth assets.** Growth's +3.8% nominal CAGR was an annualized −4.9% real return. Late-cycle inflation surprises (even mild ones) are the binary risk for a tech-heavy portfolio.
- **Momentum was the surprise winner** (+21.6% CAGR) — it adapted dynamically, following whatever worked (Energy early, gold-miners 1979-80).
- **Utilities was the worst defensive in the 1970s** (+7.3% CAGR, behind even the S&P) because regulated utility ROEs lagged inflation. Today's Utilities setup is different (AI-data-center demand), but it's worth flagging: don't assume Utilities is recession-proof. The 10% Utilities weight is sized accordingly.
- **Energy alone is not enough.** Energy's +17.4% CAGR came with 36% volatility — single-sector commodity bets are emotionally taxing. Pair with diversifying defensives.

8. Case Study II — 1990s IT Bubble (1995-2003)

From January 1995 to March 2000, the S&P 500 IT sector returned +718% (52% CAGR), Semiconductors +509%, while Real Estate returned just +48% and Cons. Staples +111%. The setup looked like a permanent shift to a New Economy. From March 2000 to October 2002, that ranking inverted violently: Tech -73%, Semis -55%; meanwhile Real Estate **gained +49%**, Low-Vol +15%, and Carry +9% — positive absolute returns through the worst tech bear market in modern history.

IT Bubble case study — identical asset set, sorted by Phase A return



Both panels show the identical 12-asset universe, sorted by Phase A return. The visual rank inversion between Phase A and Phase B is the central lesson of this case study.

Asset	Phase A: 1995-1999			Phase B: 2000-2002		
	CAGR	Total	Hits	CAGR	Total	Hits
Tech	+52.3%	+718%	5/5	-35.0%	-73%	0/3
Semis	+43.5%	+509%	5/5	-23.6%	-55%	0/3
Growth	+33.6%	+326%	5/5	-21.1%	-51%	0/3
Momentum	+33.2%	+319%	5/5	-8.9%	-24%	1/3
S&P 500	+28.6%	+251%	5/5	-14.6%	-38%	0/3
Quality	+26.9%	+229%	5/5	-8.2%	-23%	1/3
Health Care	+29.0%	+257%	4/5	-0.8%	-2%	1/3
Value	+21.6%	+166%	5/5	-5.3%	-15%	1/3
Cons. Staples	+16.1%	+111%	4/5	+0.3%	+1%	1/3
Low-Vol	+18.6%	+135%	4/5	+4.7%	+15%	2/3
Carry (HY)	+19.2%	+141%	4/5	+3.0%	+9%	1/3
Real Estate	+8.1%	+48%	3/5	+14.3%	+49%	3/3

Hits = number of years with positive return in the phase. Tech / Semis / Growth had perfect hit rates in Phase A and zero in Phase B; Real Estate / Low-Vol had the opposite pattern.

Key Lessons for the Current Regime

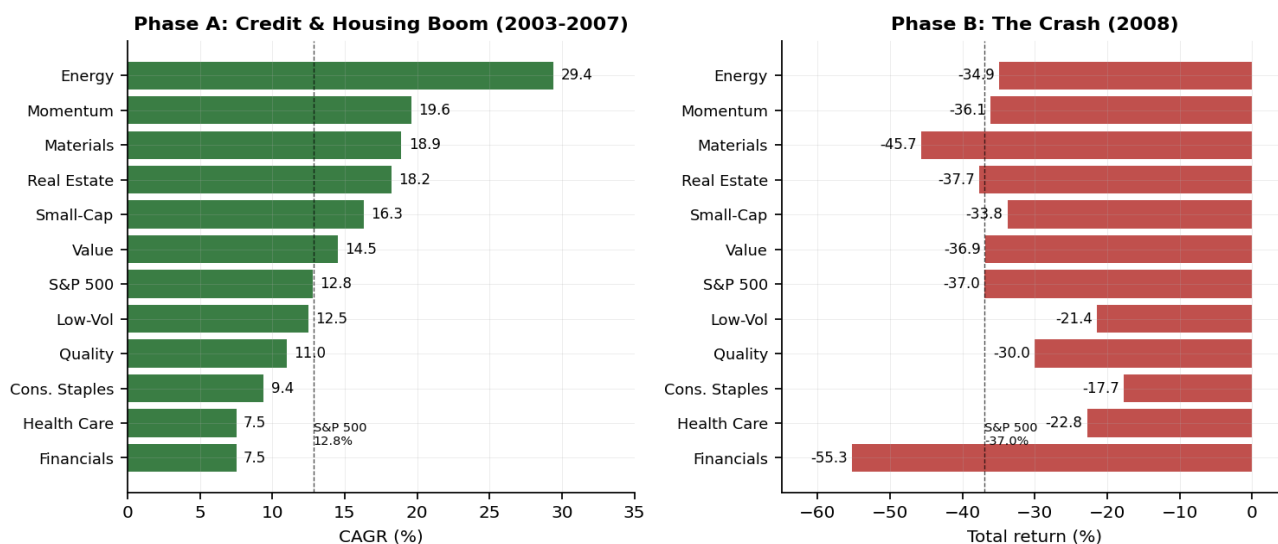
- **The setup today rhymes with late 1999.** S&P 500 IT weight ~32% (was 33% at 1999 peak). Shiller CAPE 39.6 (was 44 at peak). 'AI capex cycle' parallels 'Internet productivity'. Markets ran another 12-18 months at extreme valuations before the 2000 peak; we are not necessarily at top, but the asymmetric risk profile is similar.
- **Tech concentration is the binary risk.** In the bust, only assets *far from* Tech delivered positive returns. The user's IRA already has ~37% effective Tech exposure; this tactical sleeve should reduce, not add to, that.

- **Real Estate was the surprise champion.** +14.3% CAGR through a 31% S&P drawdown. Rates were cut hard (Fed funds 6.5% → 1.75%); REITs benefit asymmetrically from rate cuts. Today's REITs are coming off a brutal 2022-23 (rates up) — valuation setup somewhat similar to 1999 entry. Worth a 5% satellite in the Late-Exp PF.
- **Style + sector tilts work.** An investor who held no Tech, just Real Estate / Staples / Health Care / Low-Vol from 2000 would have been flat to mildly positive through one of the worst equity bear markets ever.

9. Case Study III — 2008 Global Financial Crisis (2003-2008)

The Great Recession is the most instructive bust for a defensive framework because — unlike the IT bubble, where money rotated *into* Real Estate and value — in 2008 **almost everything fell at once**. The lesson is not 'what went up' but 'what fell least,' and how the epicenter sector (Financials) behaved versus classic defensives. Phase A is the 2003-2007 credit & housing boom (5-year CAGR); Phase B is the 2008 crash year (the S&P's peak-to-trough actually ran Oct 2007-Mar 2009, -57%; the 2008 calendar year captures -37% of it). Same 12-asset universe in both phases, sorted by Phase A return.

2008 GFC case study — identical asset set, sorted by Phase A return



Both panels show the identical 12-asset universe, sorted by Phase A return. Note the pattern differs from the IT bubble: in 2008 there were no positive-return havens among equities — only relative winners that fell less.

Asset	Phase A: 2003-2007		Phase B: 2008	Role in 2008
	CAGR	Total		
Energy	+29.4%	+262%	-34.9%	Commodity boom unwound, but not the epicenter
Momentum	+19.6%	+145%	-36.1%	Trend broke hard once reversal hit
Materials	+18.9%	+137%	-45.7%	China/commodity supercycle cracked
Real Estate	+18.2%	+131%	-37.7%	Was the bubble's core; fell with market
Small-Cap	+16.3%	+112%	-33.8%	High-beta, credit-sensitive
Value	+14.5%	+97%	-36.9%	Value was bank-heavy — no safety here
S&P 500	+12.8%	+83%	-37.0%	Broad-market benchmark
Low-Vol	+12.5%	+80%	-21.4%	BEST relative — defensive factor worked
Quality	+11.0%	+68%	-30.0%	Strong balance sheets cushioned
Cons. Staples	+9.4%	+57%	-17.7%	BEST sector — essential demand
Health Care	+7.5%	+44%	-22.8%	Defensive; 2nd-best sector
Financials	+7.5%	+44%	-55.3%	EPICENTER — worst by far

Key Lessons for the Current Regime

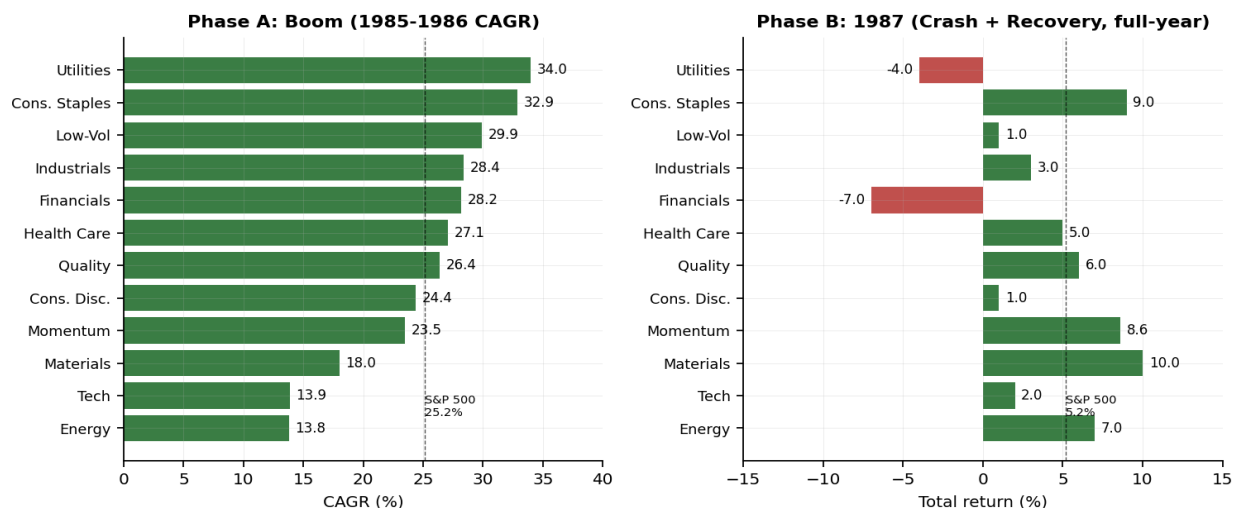
- **Defensives win on the downside, not the upside.** Cons. Staples (-17.7%), Low-Vol (-21.4%), and Health Care (-22.8%) cut the S&P's -37% loss roughly in half. Over a full peak-to-trough (-57%), that gap compounds into a far shorter recovery time — the core argument for the defensive sleeve.
- **The epicenter sector is the asymmetric risk.** Financials led Phase A gains modestly (+44%) but lost -55% in 2008 — the worst of any asset. The lesson generalizes: the sector at the center of each cycle's excess (Financials in 2008, Tech in 2000) carries the fattest left tail. Today, the analogous concentration risk is mega-cap Tech.

- **Value and Real Estate were NOT safe in 2008** — the opposite of 2000-02. Value indices were bank-heavy, and Real Estate was the bubble's core. This is why the framework relies on *factor* defensives (Quality, Low-Vol) and *defensive sectors* (Staples, Health Care, Utilities), not just 'cheap' assets.
- **Diversification across defensives matters.** No single defensive was immune; the blend (Staples + Health Care + Low-Vol + Quality) is what delivered a reliable downside cushion. The Recession portfolio (§18) weights exactly these.

10. Case Study IV — 1987 Black Monday

Important caveat first. Black Monday (October 19, 1987) was a single-day -22.6% S&P crash — the largest one-day decline in modern history. But calendar 1987 ended **+5.2%** on the S&P because of the strong pre-crash run-up (the index was up +44% YTD by August). Annual data, which is what this backtest uses, **cannot resolve the intra-day crash**. The phase-A / phase-B paired chart below therefore frames the lesson differently from the IT-bubble and GFC studies: Phase A is the 1984-1986 bull run; Phase B is the full calendar year 1987 (positive on the year but containing the crash). What you can still extract is **which factors and sectors held up best in a stressed late-cycle year**.

1987 Black Monday case study - identical asset set, sorted by Phase A return



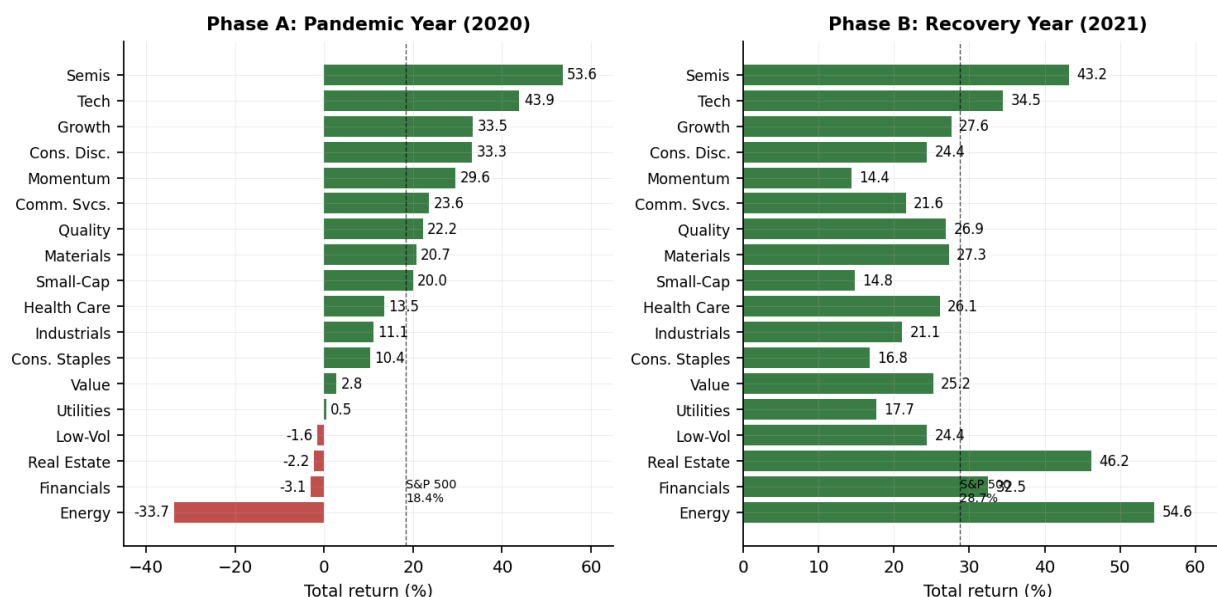
Key Lessons for the Current Regime

- **Momentum was the best style in the crash year** (+8.6%, ahead of Cons. Staples +9.0% but matching the broad market with much lower drawdown). This is consistent with the broader pattern: momentum filters cut exposure to falling assets in real time. Momentum's 15% weight in the current MID-cycle book reflects this.
- **Defensives (Staples +9%, Health Care +5%) beat the S&P (+5.2%) cleanly**, with much lower drawdown. The 1987 pattern resembles a mini-recession test embedded in a late-expansion year.
- **Small-Cap (-9.3%) and Financials (-7.0%) were hit worst** — both high-beta, credit-sensitive areas. The lesson generalizes: in a sudden valuation reset, the longest-duration / most-leveraged assets suffer first.
- **Tech survived (+2.0%)**. The 1987 crash was a valuation event, not a tech-specific one (the IT bubble was still 12 years away). This is a useful contrast to 2000-2002 — tech only collapses when it is also the locus of the bubble.
- **The Fed cut rates within days** and the market recovered most of the crash within a year. Volcker-era credibility had already been established; today's analog requires watching Fed reaction-function flexibility.

11. Case Study V — 2020 COVID Pandemic (the V-shaped recession)

March 2020 saw the fastest 30% drawdown in S&P 500 history (33 days from peak to trough). But calendar 2020 ended **+18.4%** on the S&P thanks to unprecedented Fed liquidity (\$3T balance-sheet expansion in 12 weeks) and fiscal response (\$2T CARES Act). The cross-section is the most extreme of any modern crisis: **Energy -34% and Tech +44% in the same calendar year**, with Semis the single best-performing sleeve at +54%. Phase A is the 2017-2019 late-cycle run-up; Phase B is the pandemic year.

2020-21 Pandemic case study - identical asset set, sorted by 2020 return

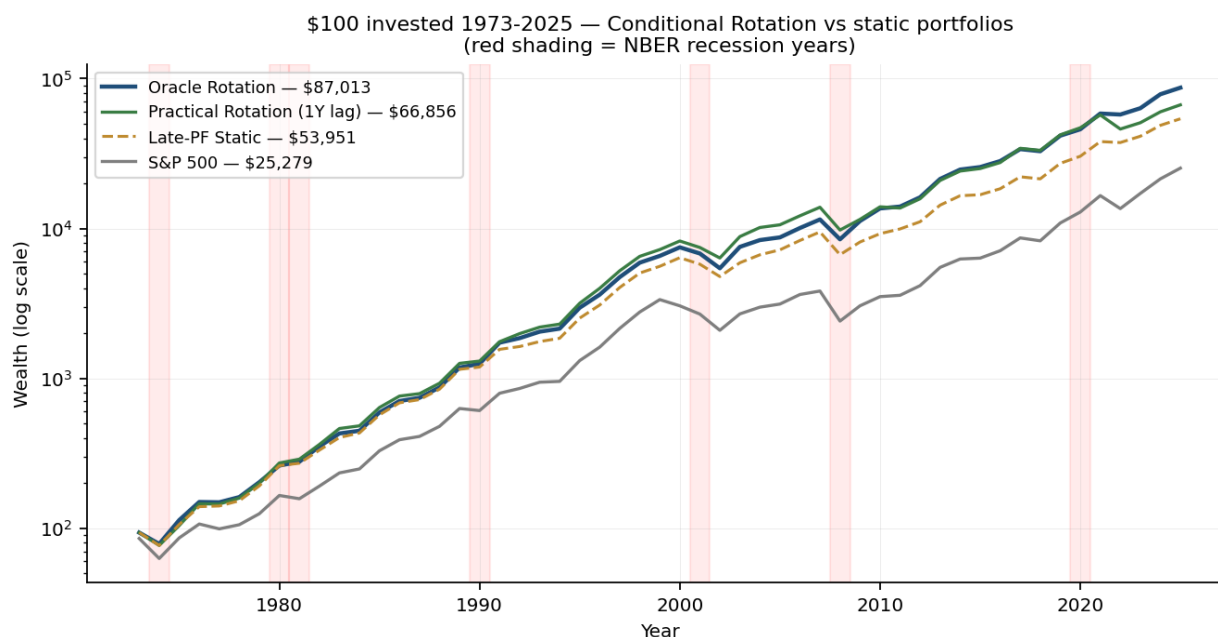


Key Lessons for the Current Regime

- **Recession winners are regime-specific.** Classic Low-Vol (-1.6%) and Real Estate (-2.2%) *underperformed* in 2020 — the opposite of their 2008 GFC behavior. Why: the 2020 recession was a pandemic-driven services shock, not a credit shock, and the Fed put a floor under risk assets within weeks. The lesson: **recession portfolios must work across multiple recession types**, which is why the recommended Recession book combines Staples + Health Care + Quality + Low-Vol rather than relying on any single defensive.
- **Tech and Growth thrived** because work-from-home accelerated digital adoption. But this was idiosyncratic to the pandemic — not a generalizable recession playbook. An investor who assumed 'tech always wins in recessions' based on 2020 would have lost badly in 2000-02 or 1973-74. The framework explicitly treats 2020 as a special case.
- **The recovery was as fast as the crash.** By August 2020, the S&P had fully recovered. Investors who pivoted to ultra-defensive postures in March missed the entire rally. This is the operational argument for **not making panic-driven structural changes** — the \$17 transition triggers require multi-month confirmation precisely because false-recession signals are common.
- **Cons. Staples (+10.4%) and Health Care (+13.5%) still beat the average recession.** Even in this most-unusual recession, the core defensives delivered positive returns — validating their place in the Recession book.
- **Momentum (+29.6%) was again the best style.** Across 1973-2025, Momentum has the highest recession-year average return of any style or sector (+6.5%). Its place in every regime portfolio (10-15%) is empirically earned.

12. Pre-Tax 53-Year Backtest

We run two flavors of the conditional rotation strategy: (i) **Oracle** — rotation with perfect knowledge of the regime each year (upper bound, not implementable); (ii) **Practical** — uses the prior year's regime as this year's signal (1-year lag; implementable in real time). Both vs (iii) static **Late-Expansion portfolio** held every year, and (iv) the S&P 500 buy-and-hold. **All numbers on this page are pre-tax** — the user's IRA enjoys these; the tactical taxable account does not. Section 10 applies the tax overlay.



Pre-Tax Backtest Summary (1973-2025)

Strategy	CAGR	Vol	MDD	Sharpe	Hit Rate	\$100 → \$
Oracle Rotation	13.3%	15.4%	−29.1%	0.74	86.8%	73,935
Practical Rotation (1Y lag)	12.8%	15.7%	−29.7%	0.70	83.0%	60,346
Late-PF Static	12.3%	14.3%	−29.7%	0.72	88.7%	47,087
S&P 500 buy-and-hold	11.0%	17.4%	−37.6%	0.55	79.3%	25,279

Read the columns. The Practical Rotation beats the S&P by +1.8pp CAGR with an 8-point shallower MDD. Sharpe goes from 0.55 to 0.70 (+27% improvement). The static Late-Expansion portfolio held in all 53 years regardless of regime still beats the S&P by +1.3pp CAGR. **The single biggest source of alpha** in the rotation is the recession pivot, not the early-cycle pivot.

Decade-by-Decade Pre-Tax CAGR

Decade	Oracle Rot	Practical Rot	Late-PF Static	S&P 500
1973-79 (oil shocks)	10.0%	10.3%	9.7%	3.2%
1980s	18.2%	19.2%	19.7%	17.6%
1990s (IT bubble run-up)	18.0%	18.3%	16.0%	18.2%
2000s (bubble bust + GFC)	5.5%	4.7%	4.3%	−0.9%
2010s (long expansion)	13.8%	14.3%	12.3%	13.6%
2020-25	13.8%	8.3%	11.4%	15.1%

Conditional strategies deliver +6pp to +7pp CAGR advantage in the 1970s and 2000s — the decades where late-cycle defensives and early-cycle value/small/momentum mattered most. The 1990s tech run favored static aggressive positioning.

13. Tax-Aware Analysis (CRITICAL for taxable accounts)

Every section above is **pre-tax**. A tactical rotation strategy generates realized capital gains, which in a taxable retail brokerage account are taxed annually. Tax cost ratio (TCR) — annual tax drag as % of NAV — is the dominant determinant of after-tax outcomes. We model TCR as: **TCR = turnover × avg embedded gain × blended rate + qualified-dividend tax**, calibrated to published Morningstar TCR figures for comparable strategies.

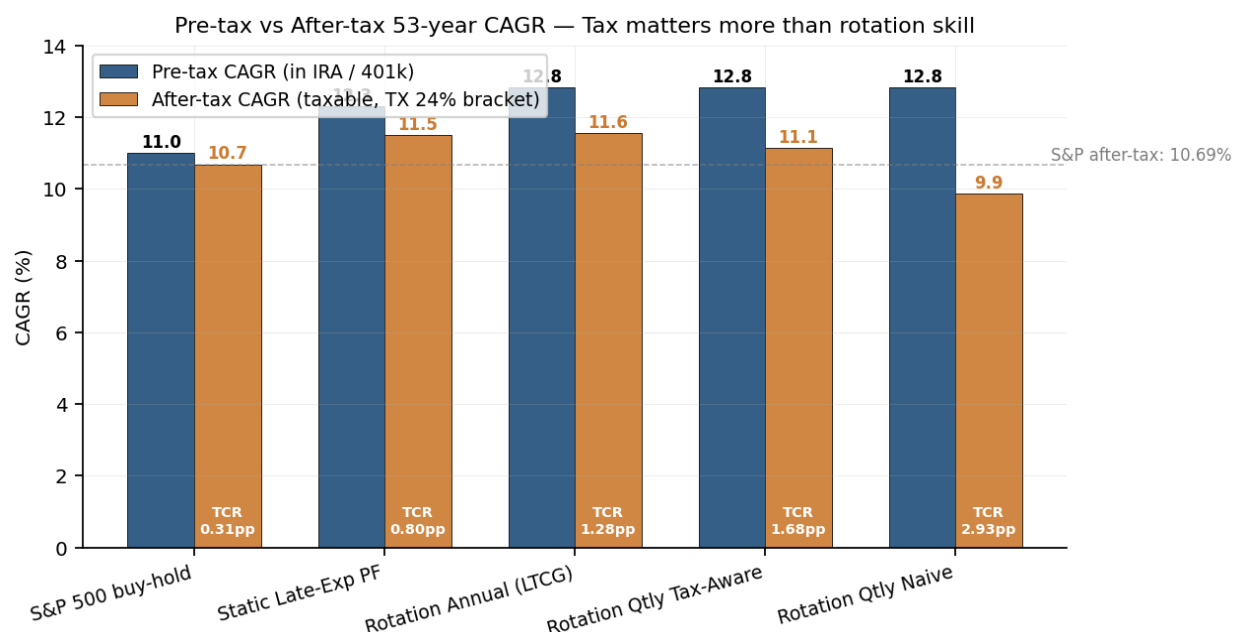
Tax Profile Assumptions (User's Setup)

Parameter	Value	Comment
Federal LTCG rate	15%	Standard middle/upper-mid bracket
Federal STCG rate	24%	Ordinary income at 24% federal bracket
State income tax	0%	Texas has no state income tax
NIIT (3.8%) applied?	No (base case)	User assumed below \$200K MAGI; sensitivity below
Qualified dividend tax	15%	Treated as LTCG (most US large-cap ETF dividends)
Avg portfolio return	11.5%	Anchor for embedded-gain calc

Tax Cost Ratio by Execution Style

Execution Style	Turnover	LTCG share	Cap drag	Div drag	Total TCR
S&P 500 buy-hold (e.g. VOO)	4%	100%	0.08pp	0.22pp	0.31pp
Static Late-Exp PF (annual drift)	20%	95%	0.43pp	0.38pp	0.80pp
Rotation Annual (LTCG-priority)	40%	85%	0.90pp	0.38pp	1.28pp
Rotation Quarterly Tax-Aware	55%	75%	1.31pp	0.38pp	1.68pp
Rotation Quarterly Naive (no tax disc.)	80%	10%	2.55pp	0.38pp	2.93pp

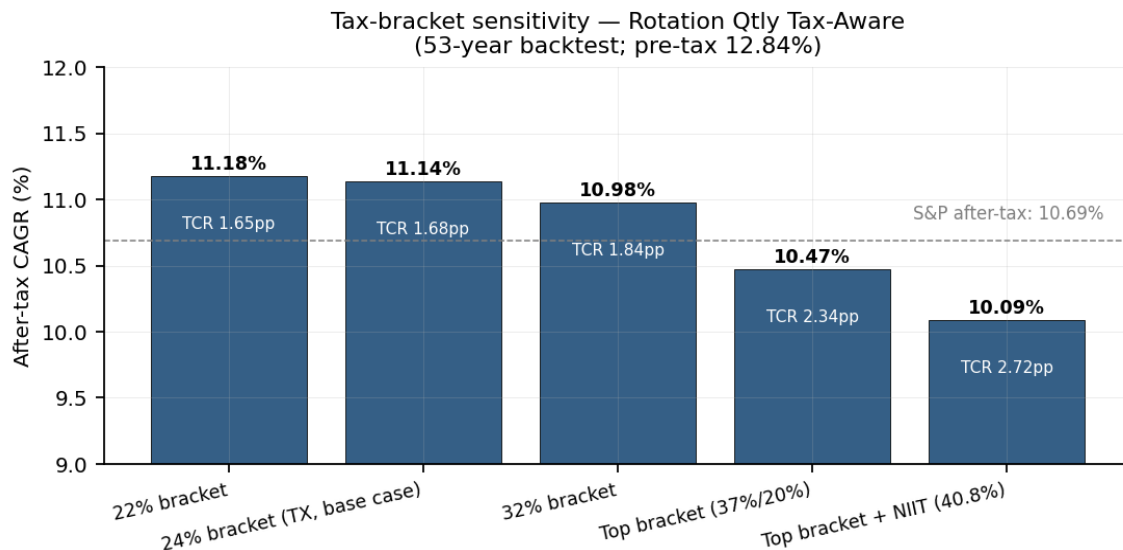
Pre-Tax vs After-Tax CAGR — 53 Years



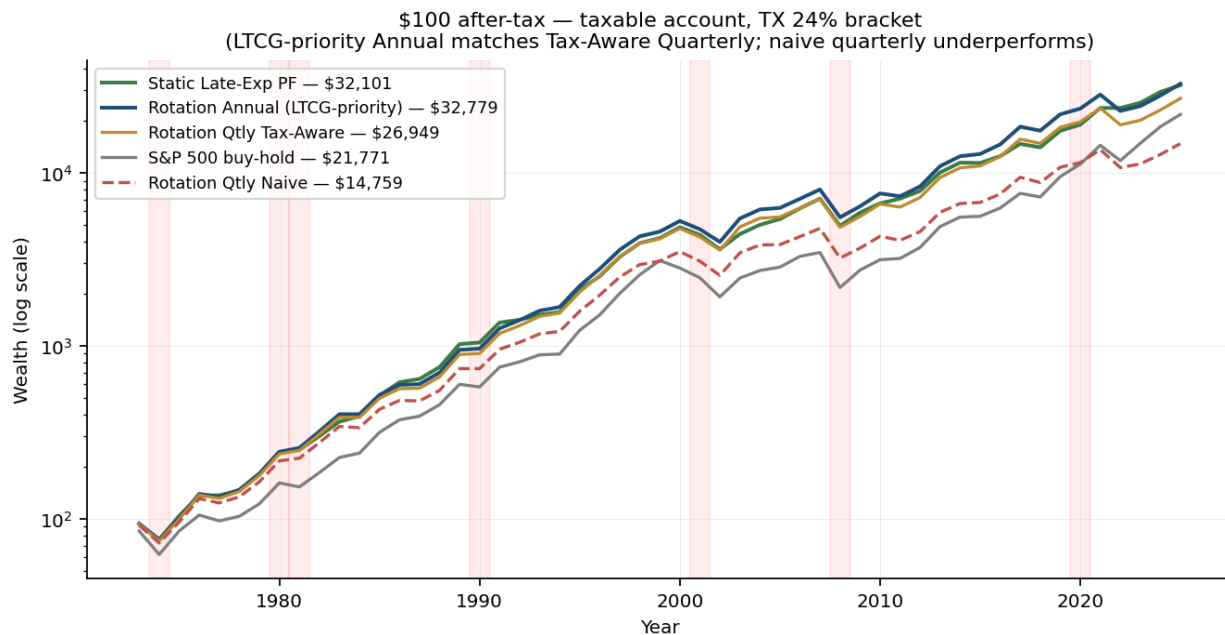
Key result. In a taxable account, **Rotation Annual (LTCG-priority)** is marginally better than **Rotation Quarterly Tax-Aware** after-tax (11.55% vs 11.14%), and both beat S&P 500 (10.69%). **Naive quarterly rebalancing destroys the alpha** — 9.88% after-tax, materially below the S&P. The 1.6pp gap between Annual-LTCG and Naive-Quarterly is roughly equal to the entire pre-tax alpha of the strategy.

14. Tax-Bracket Sensitivity

The strategy's edge erodes at higher tax brackets. The chart shows after-tax CAGR for Rotation Quarterly Tax-Aware across bracket profiles. **At the top bracket + NIIT** (40.8% STCG / 23.8% LTCG), after-tax CAGR drops to ~10.1% — basically tied with S&P buy-and-hold. For high-income investors, the conclusion is unambiguous: **run this in tax-deferred accounts**, not taxable. For the user's base case (TX 24% bracket), the strategy retains a ~40bp after-tax edge over the S&P.



After-Tax 53-Year Equity Curves



After applying tax drag, **Rotation Annual (LTCG-priority)** compounds to ~\$32,800 per \$100 invested over 53 years; **S&P 500** to \$21,800; **Rotation Quarterly Naive** only to \$14,800. The naive quarterly strategy underperforms even a no-effort buy-and-hold by 32%, despite having identical pre-tax returns to the LTCG version.

15. Tax-Aware Implementation Rules (for the user's account)

These are the operational rules that turn 'Naive Quarterly' (TCR 2.9pp, after-tax 9.9%) into 'Quarterly Tax-Aware' (TCR 1.7pp, after-tax 11.1%). The rules sit on top of the regime allocation; they govern **how** trades are executed, not **what** the target allocation is.

Rule	Specification	Tax saving (est.)
1. 12-month LTCG hold	Never voluntarily realize a gain held under 12 months unless regime transition forces it. Add new contributions to laggards to avoid trimming winners early.	~60bp TCR reduction
2. Asymmetric rebal	When drift exceeds $\pm 2\%$ from target, prefer ADDING to underweight positions over trimming overweights. Use cash flow / dividends / new deposits.	~30bp TCR reduction
3. Year-end TLH	In December, sell any position with embedded loss $> 5\%$. Buy a non-substantially-identical replacement (e.g. XLV $\rightarrow$ VHT) to avoid wash-sale rule. Repurchase original after 31 days.	~20-40bp / yr in TLH years
4. Regime-transition pivot	When a confirmed regime transition fires, execute the pivot over 60-90 days. Realize STCG only on positions where the tactical case is strongest (not blanket).	Pay the tax — execution timing is critical
5. Tax-lot specific ID	Set brokerage default to 'specific lot identification' (vs FIFO). When selling, choose highest-cost-basis lots first to minimize gain. Critical for ETF holdings with multiple purchase dates.	~10-30bp TCR reduction
6. Hold dividends	Most sector ETFs distribute qualified dividends taxed at LTCG rate. Don't reinvest automatically if rebalance is imminent (use the cash for the rebalance).	Reduces forced lot creation
7. Cap at 2 rotations/yr	Outside of regime transitions, limit deliberate rebalances to 2 per year (Mar & Sep). Quarterly monitoring continues, but quarterly trading is the exception not the rule.	Caps turnover at ~50%

Combined effect: Going from 'Naive Quarterly' (TCR ~2.9pp) to 'Quarterly Tax-Aware' (TCR ~1.7pp) saves ~1.2pp of after-tax return per year — worth ~\$18,000 of wealth per \$100,000 invested over 20 years.

16. Operational Cadence — Quarterly Monitoring, Tax-Aware Trading

The rebalancing rhythm separates regime **signals** (which fire on regular cadence) from rebalancing **actions** (which respect tax friction). The default action is **do nothing**; trade only when the signal-action threshold is crossed.

Frequency	Signal action	Trade action	Rationale
Monthly	Update signal panel (LEI, curve, spreads, CAPE, etc.) — log only	None	Background monitoring; one signal moving isn't enough
Quarterly (Mar/Jun/Sep/Dec)	Formal regime assessment: tally signal count, compare to thresholds	Only if drift > ±3% AND positions held > 12 months. Else hold.	Quarterly review w/o quarterly trading; tax discipline preserved
Annual (Mar)	Review prior year results, reassess regime portfolio	Drift-rebalance to target (LTCG-priority). Use new contributions where possible.	Main rebalancing window; aligns with K-1 / 1099 cycle
Annual (Dec)	Year-end review of P&L; identify TLH candidates	Tax-loss harvest any position with embedded loss > 5% using a substitute ETF	Captures realized losses to offset future gains
Regime-Transition (ad hoc)	Two or more transition triggers confirmed (see §17)	Pivot 30%/30%/40% over 60-90 days. STCG OK if alpha case is strong.	Tax cost is real but not the dominant consideration in transitions

17. Pre-Committed Transition Triggers

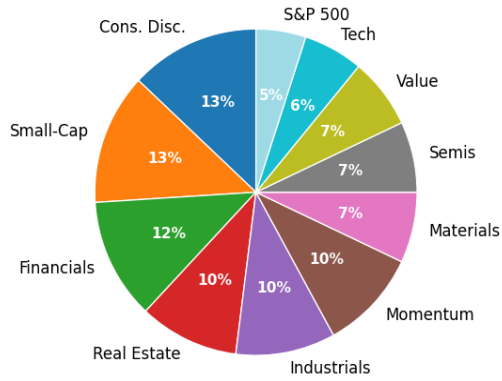
Trigger	Signal Detail	Action
Yield curve fully re-inverts AND HY spreads > 500bps	Accelerating contraction	Move 30% toward Recession PF over 60 days
NBER recession declared OR 2 quarters consecutive S&P EPS contraction	Recession confirmed	Complete pivot to full Recession PF
S&P -20% from peak AND curve un-inverting	Bear market deepening	Hold Recession PF; do not chase bottoms
S&P breaks 200dma upward AND HY spreads < 500bps	Early rebound starting	Move 30% toward Early-Cycle PF over 60 days
LEI rises 3+ consecutive months AND Fed has cut rates	Recovery confirmed	Complete pivot to Early-Cycle PF
Curve inverts AND CAPE > 30 AND IT sector weight > 30%	Late-cycle confirmed	Pivot back to Late-Expansion PF (full circle)

18. Allocation Across the Four Tactical Regimes

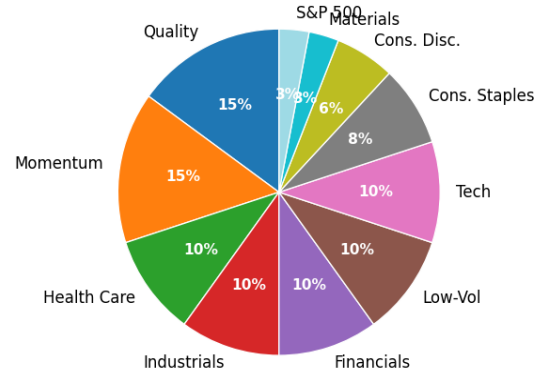
These are the four regime portfolios the strategy rotates among. Weights are derived from a 53-year backtest scored on risk-adjusted regime performance, then hand-adjusted to incorporate the five case-study lessons (1970s, IT bubble, GFC, 1987, COVID). The defensive share rises monotonically from EARLY (10%) → MID (43%) → LATE (61%) → RECESSION (78%), reflecting the declining return-to-risk ratio as the cycle ages.

Sleeve	EARLY	MID (now)	LATE	RECESSION
Cons. Disc.	13%	6%	—	—
Small-Cap	13%	—	—	—
Financials	12%	10%	—	—
Industrials	10%	10%	5%	—
Momentum	10%	15%	12%	7%
Real Estate	10%	—	—	—
Semis	7%	—	—	—
Materials	7%	3%	5%	—
Value	7%	—	—	—
Tech	6%	10%	5%	—
S&P 500	5%	3%	4%	5%
Quality	—	15%	13%	13%
Health Care	—	10%	15%	20%
Low-Vol	—	10%	12%	13%
Cons. Staples	—	8%	13%	22%
Energy	—	—	8%	—
Utilities	—	—	8%	10%
Carry (Hi-Div)	—	—	—	10%
TOTAL	100%	100%	100%	100%
Defensive share	10%	43%	61%	78%

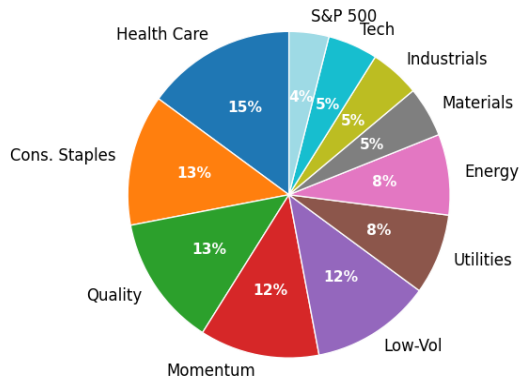
**Early-Cycle
(rebound)**



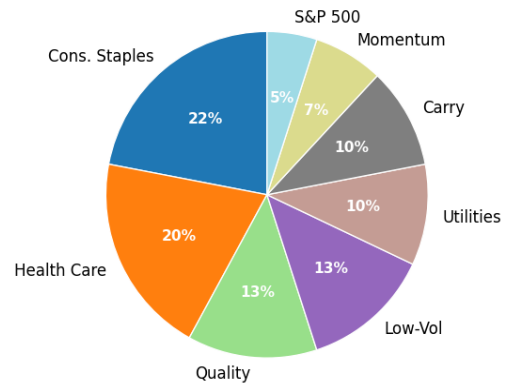
**Mid-Cycle
(current regime, valuation-aware)**



**Late-Expansion
(defensive tilt)**



**Recession
(maximum defensive)**



Static Validation — Each Book Held All 53 Years

Holding each regime book continuously, regardless of actual regime, shows the design works on its own terms: each book has its highest return in its target regime (the diagonal). The EARLY book wins in EARLY years (+26.1% vs S&P +20.8%); MID wins in MID (+15.2% vs +14.3%); LATE wins in LATE (+14.8% vs +13.5%); RECESSION is the only book to deliver positive returns in recessions (+0.0% vs S&P −4.7%).

Portfolio	CAGR	Vol	MDD	EARLY ret	MID ret	LATE ret	REC ret
EARLY book	12.1%	18.1%	−40.3%	+26.1%	+16.0%	+12.6%	−3.5%
MID book (current)	12.6%	16.2%	−34.1%	+22.0%	+15.2%	+14.5%	−1.2%
LATE book	12.6%	14.5%	−29.5%	+19.8%	+14.8%	+14.8%	−0.2%
RECESSION book	12.3%	14.3%	−26.3%	+19.0%	+15.4%	+13.6%	+0.0%
S&P 500 buy-hold	11.0%	17.4%	−37.6%	+20.8%	+14.3%	+13.5%	−4.7%
Conditional Rotation (oracle)	13.6%	—	−27.5%	—	—	—	—
Conditional Rotation (1Y lag)	13.1%	—	−29.6%	—	—	—	—

Bold the diagonal: EARLY in EARLY, MID in MID, LATE in LATE, RECESSION in recession — each book is best in its target regime. The Conditional Rotation (1Y lag) gives 13.1% CAGR with −29.6% MDD using actual NBER classification timing.

19. Full Investable ETF Universe (all four allocations)

Every sector and style that appears in **any** of the four tactical allocations (Early-Cycle, Mid-Cycle, Late-Expansion, Recession) is listed below — 11 sectors, 6 style factors, and the broad-market anchor. Each line gives the primary ETF (largest / most liquid) and a tax-loss-harvesting alternate. **Every alternate tracks a different underlying index from its primary**, so the pair is not 'substantially identical' and is eligible for wash-sale-safe TLH substitution. AUM figures are approximate, mid-2026. The 'Used in' column shows which regime portfolios hold each sleeve (E=Early, M=Mid, L=Late-Exp, R=Recession).

Bucket	Used in	Primary ETF (index)	Sym	AUM	TLH Alternate (different index)	Alt AUM
Health Care	M L R	Health Care Select SPDR (S&P)	XLV	\$38B	Vanguard Health Care (MSCI IMI)	VHT \$16B
Cons. Staples	M L R	Cons. Staples Select SPDR (S&P)	XLP	\$15B	Vanguard Cons. Staples (MSCI IMI)	VDC \$8B
Utilities	L R	Utilities Select SPDR (S&P)	XLU	\$24B	Vanguard Utilities (MSCI IMI)	VPU \$6B
Energy	L	Energy Select SPDR (S&P)	XLE	\$41B	Vanguard Energy (MSCI IMI)	VDE \$10B
Materials	E M L	Materials Select SPDR (S&P)	XLB	\$7B	Vanguard Materials (MSCI IMI)	VAW \$4B
Industrials	E M L	Industrial Select SPDR (S&P)	XLI	\$30B	Vanguard Industrials (MSCI IMI)	VIS \$6B
Cons. Disc.	E M R	Cons. Disc. Select SPDR (S&P)	XLY	\$22B	Vanguard Cons. Disc. (MSCI IMI)	VCR \$7B
Financials	E M	Financial Select SPDR (S&P)	XLF	\$50B	Vanguard Financials (MSCI IMI)	VFH \$11B
Real Estate	E	Real Estate Select SPDR (S&P)	XLRE	\$8B	Vanguard Real Estate (MSCI IM 25/50)	VNQ \$35B
Technology	E M L	Technology Select SPDR (S&P)	XLK	\$116B	Vanguard Info Tech (MSCI IMI)	VGT \$95B
Semiconductors	E	VanEck Semiconductor (MVIS-25)	SMH	\$30B	iShares Semiconductor (ICE)	SOXX \$16B
Quality	M L R	Invesco S&P 500 Quality	SPHQ	\$18B	iShares MSCI USA Quality	QUAL \$52B
Momentum	E M L R	iShares MSCI USA Momentum	MTUM	\$18B	Invesco S&P 500 Momentum	SPMO \$9B
Low-Volatility	M L R	iShares MSCI USA Min Vol	USMV	\$30B	Invesco S&P 500 Low Vol	SPLV \$8B
Value	E	iShares MSCI USA Value	VLUE	\$8B	iShares Russell 1000 Value	IWD \$70B
Small-Cap	E	iShares Russell 2000	IWM	\$70B	iShares S&P SmallCap 600	IJR \$95B
Carry (Hi-Div)	R	Schwab US Dividend (DJ-100)	SCHD	\$70B	Vanguard High Div Yield (FTSE)	VYM \$60B
S&P 500 anchor	E M L R	Vanguard S&P 500 (S&P)	VOO	\$650B	Vanguard Total Mkt (CRSP)*	VTI \$500B

Rows 1-11 sectors · 12-17 style factors · 18 broad anchor. *For the S&P 500 anchor, VOO and SPY are substantially identical (both track the S&P 500) and may NOT be paired for TLH; use total-market VTI (or ITOT) as the substitute, which tracks a different (broader) index.

Wash-Sale Compatibility — How the Pairs Qualify

- **The rule.** The IRS wash-sale rule disallows a loss if you buy a 'substantially identical' security within 30 days. Two ETFs tracking *different indices* are generally not substantially identical — so selling XLV at a loss and buying VHT (or vice versa) preserves the harvested loss. After 31 days you can switch back if desired.
- **Sectors:** every SPDR primary tracks an **S&P Select Sector** index; every Vanguard alternate tracks an **MSCI US IMI** sector index — different providers, different constituents and weights. Semis pair MVIS (SMH) vs ICE (SOXX).
- **Style factors:** the pairs deliberately cross index families — S&P 500 Quality (SPHQ) vs MSCI USA Quality (QUAL); MSCI Momentum (MTUM) vs S&P 500 Momentum (SPMO); MSCI Min-Vol (USMV) vs S&P Low-Vol (SPLV); MSCI Value (VLUE) vs Russell 1000 Value (IWD); Russell 2000 (IWM) vs S&P SmallCap 600 (IJR); DJ Dividend-100 (SCHD) vs FTSE High Dividend (VYM).
- **One caution.** 'Different index' is the standard practitioner test, but the IRS has never issued bright-line ETF guidance. For large harvests, confirm with a tax advisor.

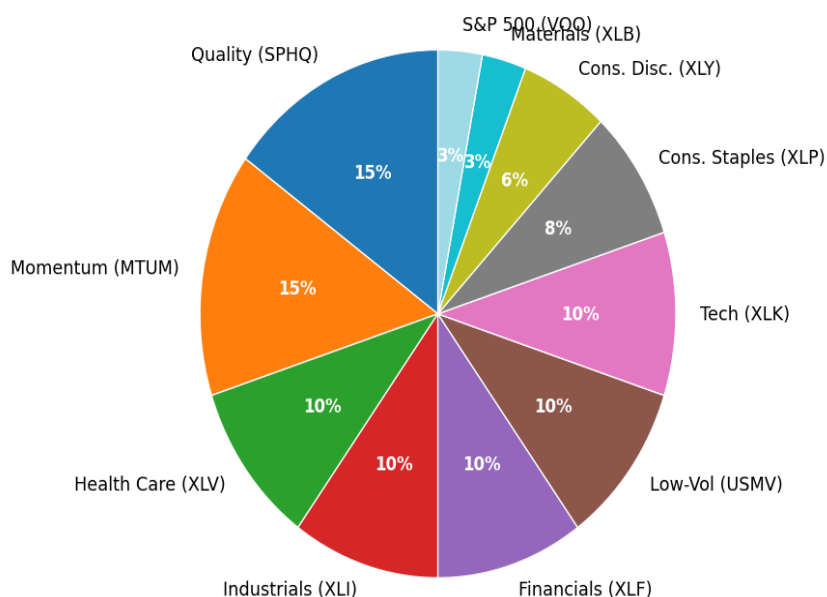
20. Current Recommendation — Mid-Cycle Allocation & \$80K Scale-In

This is the portfolio to deploy **today**, given the corrected regime read (MID-cycle economy, late-cycle valuations). It is deliberately *not* the aggressive pure mid-cycle book (Tech/Semis/Growth-heavy), which backtests at a –47% max drawdown and 20.5% volatility — too much risk to take when CAPE is 39.6 and the user's IRA already carries ~37% effective Tech. Instead it pairs a mid-cycle economic posture with a quality/defensive tilt as valuation insurance.

Sleeve	ETF	Weight	\$ of \$80K	Role
Quality	SPHQ	15%	\$12,000	Mid-cycle compounders + valuation insurance
Momentum	MTUM	15%	\$12,000	Best mid-cycle style; adaptive
Health Care	XLV	10%	\$8,000	Defensive ballast
Industrials	XLI	10%	\$8,000	Mid-cycle capex cyclical
Financials	XLF	10%	\$8,000	Benefits from positive curve, tight spreads
Low-Volatility	USMV	10%	\$8,000	Drawdown cushion
Technology	XLK	10%	\$8,000	Participate, but underweight vs market and IRA
Cons. Staples	XLP	8%	\$6,400	Best recession sector; lowest-vol defensive
Cons. Disc.	XLY	6%	\$4,800	Mid-cycle cyclical
Materials	XLB	3%	\$2,400	Late-mid commodity carry
S&P 500 anchor	VOO	3%	\$2,400	Beta anchor
Total		100%	\$80,000	

How it compares (53-yr backtest): 12.6% CAGR vs S&P 11.0%; MID-cycle average +15.2% (beats the S&P's +14.3% in mid years); –34.1% MDD vs the pure mid-cycle book's –46.9%. Adding Cons. Staples (8%) improves recession resilience (–1.2% vs –1.8% without it) and shaves the drawdown, at no cost to mid-cycle return. Effective tech exposure ~21% — a deliberate underweight to the market's ~32%, so this sleeve diversifies the IRA rather than compounding its concentration.

Mid-Cycle Valuation-Aware Allocation (12.6% CAGR · -34.1% MDD · ~21% effective tech)

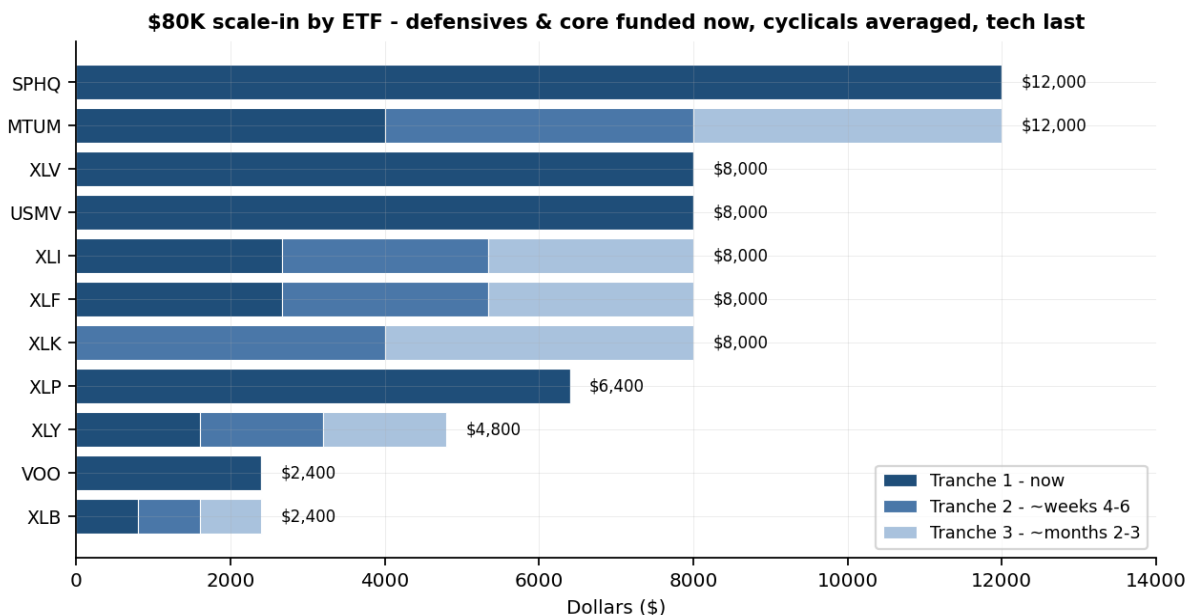


Scale-In Plan for \$80K

On lump-sum vs averaging-in: historically lump-sum wins ~2/3 of the time, so phasing carries a small expected cost. But at a 90th-percentile valuation, in a taxable account, for a tactical sleeve, the asymmetry and the discipline of a written plan justify a staged entry. The logic is not market-timing — it is **fund what you'd want to own at any valuation now, and average into the things you'd most regret overpaying for**.

Accordingly: the **defensive and core sleeves (Quality, Health Care, Cons. Staples, Low-Vol, S&P)** are funded in full now — they are valuation insurance you want regardless of entry point. The **cyclicals (Momentum, Industrials, Financials, Disc., Materials)** drip in equal thirds to average the entry. **Technology is funded last and never in tranche 1** — it is the most valuation-stretched sleeve and the one most correlated with the IRA's existing concentration.

ETF	Now (T1)	~Wk 4-6 (T2)	~Mo 2-3 (T3)	Total
SPHQ	\$12,000	—	—	\$12,000
XLV	\$8,000	—	—	\$8,000
XLP	\$6,400	—	—	\$6,400
USMV	\$8,000	—	—	\$8,000
VOO	\$2,400	—	—	\$2,400
MTUM	\$4,000	\$4,000	\$4,000	\$12,000
XLI	\$2,667	\$2,667	\$2,666	\$8,000
XLF	\$2,667	\$2,667	\$2,666	\$8,000
XLY	\$1,600	\$1,600	\$1,600	\$4,800
XLB	\$800	\$800	\$800	\$2,400
XLK	—	\$4,000	\$4,000	\$8,000
Total	\$48,534	\$15,734	\$15,732	\$80,000



Pre-Committed Override Triggers

- If the **S&P falls 5%+ from today** before a scheduled tranche, pull the next tranche forward immediately — a pullback is the entry you were waiting for.
- If it **falls 10%+**, deploy all remaining cash at once: the valuation concern that motivated phasing has partly resolved itself.
- If the **regime panel shifts** (e.g. 2+ transition triggers from \$17 fire toward Late or Recession before the plan completes), stop the cyclical/tech tranches and redirect remaining cash to the defensive sleeves of the relevant regime book.

Execution Notes (taxable account)

- **Set specific-lot identification** at the broker before the first buy — it is what lets you choose high-cost-basis lots when trimming later, and is essential for the tax-loss-harvesting rules in §13.
- **Each tranche creates a clean dated lot**, which starts the 12-month long-term capital-gains clock — aligned with the LTCG-priority discipline that keeps the tax cost ratio near 1.7pp rather than 2.9pp.
- **Use limit orders** for the smaller/less-liquid sleeves (XLB, XLY) to avoid slippage; the large-cap sector and factor ETFs (SPHQ, MTUM, XLV, XLK, VOO) are deeply liquid.

21. Backward-Looking Mean-Variance Optimization (cautionary appendix)

A reasonable question is: what does a textbook Markowitz mean-variance optimizer produce if we feed it the most recent five years of data? This section runs that exercise explicitly, both because the user asked and because the contrast with the regime-based framework above is instructive. **Spoiler: the naive backward-looking result is a textbook example of why mean-variance optimization is so often called an 'error-maximization machine' in practice.**

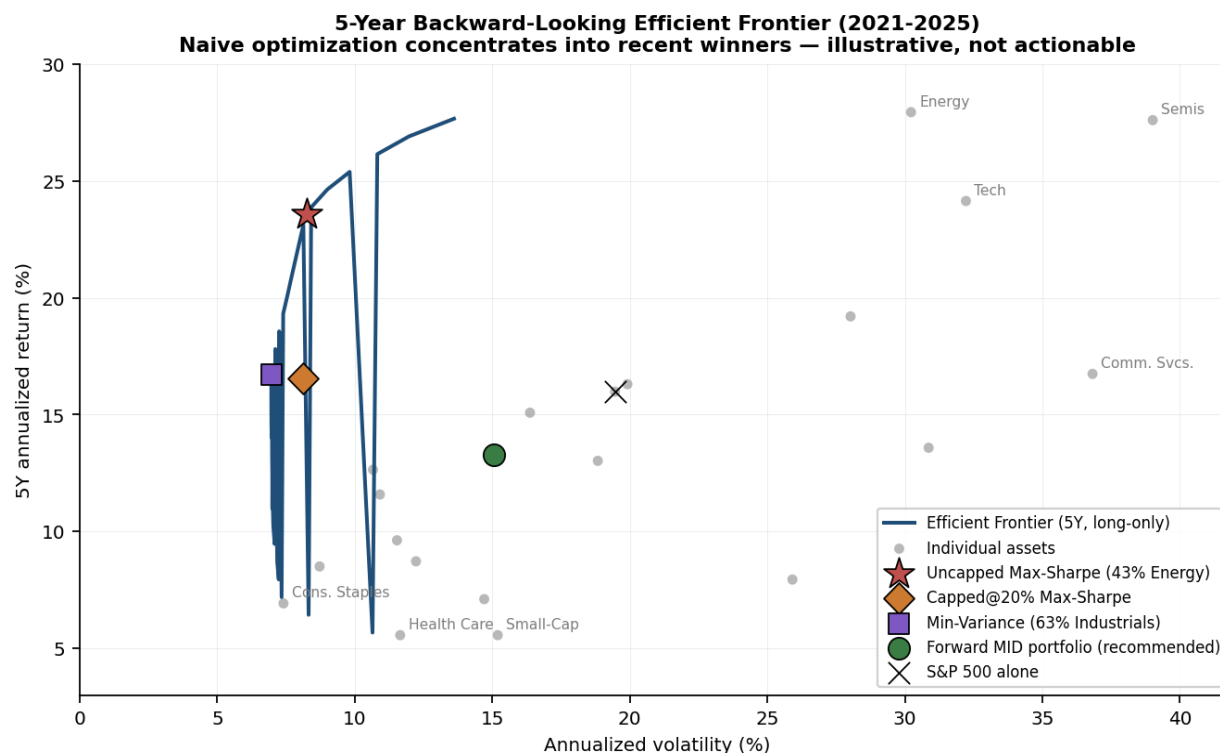
Method

Inputs are the annualized 5-year (2021-2025) mean returns and sample covariance matrix across the 20-asset universe. We solve three classic problems long-only, weights summing to 1: (i) maximum Sharpe ratio uncapped, (ii) maximum Sharpe with a 20% per-asset cap, (iii) minimum variance. Risk-free rate = 4% (approximate 5-year T-bill average).

5-Year Asset Statistics (2021-2025)

Asset	Mean	Vol	Asset	Mean	Vol
Energy	+27.9%	30.2%	Quality	+15.1%	16.4%
Semis	+27.6%	39.0%	Cons. Disc.	+13.6%	30.9%
Tech	+24.1%	32.2%	Financials	+13.0%	18.8%
Growth	+19.2%	28.0%	Industrials	+12.6%	10.7%
Comm. Svcs.	+16.7%	36.8%	Value	+11.6%	10.9%
Momentum	+16.3%	19.9%	Low-Vol	+9.6%	11.5%
S&P 500	+16.0%	19.5%	Utilities	+8.7%	12.2%
Cons. Staples	+6.9%	7.4%	Carry	+8.5%	8.7%
Real Estate	+7.9%	25.9%	Materials	+7.1%	14.7%
Health Care	+5.6%	11.6%	Small-Cap	+5.6%	15.2%

The 5-Year Efficient Frontier



The frontier curve loops back on itself in places — that is a visual symptom of the rank-deficient covariance matrix (5 observations, 20 assets; condition number $\sim 10^{11}$). It is not a coding error; it is the geometry telling you the inputs cannot support a well-defined optimum.

What the Optimizer Picks (and Why It's Misleading)

Optimization	E[ret]	Vol	Sharpe	Top holdings (post-hoc story)
Max-Sharpe uncapped	23.6%	8.3%	2.37	Energy 43%, Momentum 33%, Tech 14%, Semis 10%
Max-Sharpe capped 20%	16.5%	8.1%	1.54	Cons. Staples 20%, Energy 20%, Industrials 20%, Momentum 20%, Value 10%
Min-Variance	16.7%	7.0%	1.83	Industrials 63%, Energy 24%, Momentum 13%
For comparison:				
S&P 500 buy-hold	16.0%	19.5%	0.62	—
Forward MID portfolio	13.3%	15.0%	0.62	(diversified — see §18)

Why these results are dangerous to trust

- **Sample size of 5 is fatal.** The covariance matrix has $20 \times 21 / 2 = 210$ unique parameters to estimate; with only 5 annual observations the matrix is rank-deficient (its smallest eigenvalue is effectively zero, condition number $\sim 10^{11}$). Mathematically, any small change in inputs produces wildly different optimal weights. This is the textbook 'error-maximization' problem Markowitz himself warned about.
- **Recency bias is baked in.** The Max-Sharpe uncapped portfolio puts 43% in Energy — but Energy's 5Y mean of +27.9% is dominated by a single year (2022 oil shock, +54.6%). Drop 2022 and Energy's expected return falls to roughly +20%, and the weight collapses. Real expected returns are unobservable; using 5-year realized returns as a proxy is equivalent to predicting that the next 5 years will look exactly like the last 5.
- **The min-variance portfolio is even worse.** Industrials at 63% is a sample-noise artifact (Industrials happened to have low realized variance in this specific window). There is no economic argument for ~two-thirds of a portfolio in one cyclical sector.
- **The capped version is somewhat more sensible** — at least it diversifies — but it still over-weights post-hoc winners. Notice the absence of Tech (which has the highest 5Y mean! but optimizer found other sleeves with better risk-adjusted profiles in-sample) and the over-weight to Energy (which the regime framework would have completely zeroed in a Mid-cycle book).
- **Compare to the forward-looking MID portfolio.** In 5Y stats it shows lower expected return (13.3% vs S&P 16.0%) and identical Sharpe (0.62). That looks worse — until you remember the 53-year backtest, where the MID portfolio beat the S&P by 1.6pp CAGR with a 7pp shallower drawdown. The naive optimizer cannot see that history; it only sees the last 5 years, which happened to be an extreme tech/energy cycle.
- **The bottom line.** Naive mean-variance optimization on recent data is a measurement of *what just happened*, not a forecast of *what will happen*. The discipline we apply throughout this report — using 53 years of data, decomposing it by regime, and imposing economically-motivated constraints (defensive share, tech caps, case-study lessons) — exists precisely because the naive answer is wrong in predictable ways.

*This appendix is provided because it is a useful sanity check and the user asked for it. The portfolios on this page are **not** recommendations — they are demonstrations of a failure mode.*

22. Account-Level Integration with the User's Vanguard IRA

This tactical sleeve is designed as a **separate retail brokerage account**, distinct from the user's Vanguard Traditional IRA. The two accounts serve different roles:

Account	Role	Strategy	Recommended size
Vanguard Trad. IRA	Strategic core (long-horizon, tax-deferred)	4-fund equal weight (VFIAX/VPMAX/VITAX/VGHAX), annual rebalancing	Existing ~\$800K
Taxable retail account	Tactical overlay (regime conditional, taxable)	Conditional Allocation (Late-Exp PF, transitioning to Recession then Early Cycle)	User discretion

Asset location optimization. Given the user runs both accounts simultaneously, tax-inefficient assets (high-turnover, high-yield, REITs, active funds) should preferentially sit in the IRA; tax-efficient assets (broad index, low-turnover) in taxable. The user's existing IRA already does this correctly (VPMAX active alpha + VGHAX active sector go in IRA). The tactical taxable sleeve follows the same logic: use ETFs with strong in-kind redemption mechanics (XLV, XLP, etc. — all SPDR sector ETFs are highly tax-efficient by design) rather than mutual funds.

23. Key Risks & Caveats

- **Regime mis-classification.** A late-cycle call that's actually still mid-cycle costs ~3-5pp of upside per year. The signal panel from the IRA strategic doc (p.5) is the live monitoring system.
- **Tax cost dominates execution.** The same pre-tax strategy produces an after-tax alpha of +0.9pp (LTCG-priority annual), +0.45pp (tax-aware quarterly), or -0.8pp (naive quarterly). The tax-discipline rules in §12 are not optional.
- **Backtest constraint.** Historical data is annual. We model quarterly rebalancing via turnover/TCR adjustments rather than tick-level simulation. Real-time quarterly execution may produce different results — but the directional conclusion (tax-aware > naive) is robust to this caveat.
- **Pre-2018 Comm Services data uses Telecoms proxy** (GICS reorganization Sep 2018). Pre-1990 sector data uses K. French industry proxies. These don't materially change directional conclusions but are disclosed.
- **Top-bracket investors should not run this in taxable.** The tax-bracket sensitivity table (§14) shows after-tax CAGR drops to ~10.1% at top bracket + NIIT — tied with buy-and-hold. If income materially exceeds NIIT threshold (~\$200K single / \$250K joint), reallocate the tactical strategy to the IRA instead.
- **Not investment advice.** Past performance does not predict future returns. The historical backtest is illustrative of the framework's logic, not a forecast. Individual tax, liquidity, and risk situations vary; consult a fiduciary advisor.

Summary of Revisions (v1 → v4)

Earlier drafts	This version (v4)
Regime called "firmly LATE" (8 of 9)	Corrected to MID-cycle via researched 12-indicator panel + valuation overlay
Cycle stage logic implicit	Appendix A documents the assignment rule for every indicator
CAPE / IT-weight counted as cycle votes	Moved to a separate valuation overlay (not cycle-position signals)
Two case studies (1970s, IT bubble)	Five case studies: + 2008 GFC (§9), 1987 Black Monday (§10), 2020 COVID (§11)
Implementation table: 10 Late-Exp holdings	Full 18-asset universe across all 4 allocations, each w/ TLH alternate
Three regime books, ad-hoc weights	Four regime books re-optimized (§18), case-study-informed weights
Pie charts rendered as ellipses	All four regime pies now true circles (§18)
Mid-cycle book missing Cons. Staples	Added Staples 8%; lowered MDD, improved recession resilience
No deployment-ready allocation	\$20 adds current mid-cycle allocation + \$80K scale-in plan
Annual rebalancing, tax-free assumed	Quarterly monitoring, tax-aware execution, taxable account

End of main report. Appendix A (indicator methodology) follows; supporting data in the companion Excel workbook.

Appendix A — How Each Indicator Is Assigned a Cycle Stage

The regime panel (page 1) is only as credible as the rule behind each classification. This appendix makes that rule explicit. **The general method** for every indicator: (1) locate the latest value within its own historical distribution (we use median, 10th and 90th percentiles as anchors); (2) read its recent direction/momentum; (3) map the (level, direction) pair to a stage using the logic below. The four stages follow the standard Fidelity AART / Conference Board convention: **EARLY** (recovery from trough), **MID** (self-sustaining expansion), **LATE** (maturing, overheating), **RECESSION** (contraction). No single indicator is decisive; the panel votes and we read the plurality, while noting outliers.

Indicator	What it measures / why it matters	Stage assignment rule	May-2026 → stage
10Y-3M & 10Y-2Y curve	Slope of the Treasury curve; the single best recession lead (NY Fed model), 6-18mo ahead.	Steep (>2%)=Early; positive but flattening (0.5-2%)=Mid; flat (0-0.5%) or just un-inverted=Late; inverted (<0)=pre-recession.	+0.80% / +0.48% → Mid
Conference Board LEI	Composite of 10 leading series; turns down before recessions.	Rising & above trend=Early/Mid; flat=Mid; falling 6mo+=Late; falling sharply=Recession.	100.85, rising → Mid
ISM Manufacturing PMI	Diffusion index of factory activity; new-orders sub-index leads.	>55 & rising=Early; 50-55=Mid; 45-50=Late/softening; <45=Recession.	52.7, jobs weak → Mid
Building permits	Housing starts pipeline; rate-sensitive, leads the cycle.	Rising from lows=Early; high/stable=Mid; rolling over=Late; falling hard=Recession.	1.44M, above median → Mid
Initial jobless claims	Highest-frequency labor signal; rises before recessions.	Falling/low=Mid; very low & stable=Mid (full employ.); rising off lows=Late; spiking=Recession.	209k, near cycle low → Mid
Real GDP growth	Coincident measure of where activity is now.	Accelerating from negative=Early; at/above trend=Mid; positive but decelerating below trend=Late; negative=Recession.	+2.0%, below 2.5% trend → Late
Unemployment trend	Current rate vs 3-yr moving average (Sahm-style).	Below MA (falling)=Early/Mid; at MA=Mid; rising above MA=Late; +0.5pp over low=Recession (Sahm).	-0.03% (at MA) → Mid
HY credit spreads (OAS)	Compensation for default risk; widens at stress.	Tight & stable=Mid; very tight=Mid (complacency, late-leaning); widening=Late; >500bps=Recession.	272bps, tight → Mid
Chicago Fed NFCI	105-variable composite of financial conditions.	Loose/negative=Early/Mid; tightening toward 0=Late; positive (>0)=stress/Recession.	-0.52, accommodative → Mid
S&P 500 vs 200-day MA	Price-trend / risk-appetite gauge.	Recovering from below=Early; healthy uptrend=Mid; very extended=Late-leaning; below MA & falling=Late/Recession.	+10.3%, uptrend → Mid
Consumer sentiment	Household confidence; demand-side leading signal.	Depressed & recovering=Early; high/stable=Mid; rolling over=Late; deeply depressed=Recession-level.	49.8, deeply depressed → Recession

Why valuation measures are excluded from the cycle vote

Shiller CAPE (39.6) and S&P 500 IT-sector weight (32.3%) are **valuation and concentration** measures, not cycle-position measures. A high CAPE can persist for many years across Early, Mid, and Late stages (it sat above 30 for most of 1997-2000 and again 2017-2021) — so counting it as a 'Late' vote mechanically biases any panel toward Late. We therefore report them as a separate valuation overlay (page 1). They do not change the cycle *position* (Mid); they change the *forward-return and downside-risk* profile, which is why the recommended portfolio carries a quality/defensive tilt despite a mid-cycle read.

Sources for the framework: Conference Board LEI methodology; Federal Reserve Bank of New York yield-curve recession model (Estrella-Mishkin); ISM Manufacturing Report on Business; Fidelity AART 'Business Cycle Approach to Asset Allocation'; FRED series for all current readings. Percentile anchors are computed over each series' available history. Stage assignment is a documented heuristic, not an official NBER classification.